ECA1803

OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO1 Engineering Design Assignment Report

Design of a Standards-Compliant Optical Fiber Telecommunication Link

From Architectural Specifications to Field Deployment

60 km	1550 nm	10 → 100 Gb/s	21 dB
Link span	C-band wavelength	Initial / upgrade rate	Output OSNR (req. 17 dB)

Candidate Name	K Hemanth Kumar
Register Number	192312675L
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1. Executive Summary & Design Requirement Statement

This report describes the physical-layer design of a 60 km point-to-point optical fiber link between a metropolitan Central Office (CO) and an enterprise premises. The proposed span operates at 1550 nm in the C-band and is intended to provide an initial capacity of 10 Gb/s using Intensity Modulation and Direct Detection (IM/DD). The design also considers a later transition to 100 Gb/s coherent transmission while retaining the installed cable route.

The design is developed by examining the conditions for optical guidance, comparing fiber types defined by ITU-T, calculating the link power budget, and assessing dispersion, polarization effects, and amplifier noise. Dispersion-Compensating Fiber (DCF) and Erbium-Doped Fiber Amplifier (EDFA) operation are included in the analysis to evaluate pulse quality, Optical Signal-to-Noise Ratio (OSNR), and the required Bit Error Rate (BER).

Principal Design Conditions

- **Transmission span and wavelength:** The optical path extends for 60 km and uses a nominal operating wavelength of 1550 nm.
- **Data capacity:** The initial line rate is 10 Gb/s, giving a 100 ps bit interval. The design also examines a 100 Gb/s upgrade, for which the bit interval is 10 ps, without changing the physical route.
- **Fiber arrangement:** ITU-T G.652.D single-mode fiber forms the backbone. ITU-T G.657.A bend-insensitive fiber is used near the building entrance, with the specified bend-related excess loss limited to 0.10 dB.
- **Dispersion control:** Accumulated group-velocity dispersion is addressed using a suitably sized DCF section to reduce pulse spreading and the resulting inter-symbol interference (ISI).
- **PMD condition:** Differential group delay is limited to 10% of the bit interval, corresponding to 10 ps at 10 Gb/s and 1 ps at 100 Gb/s.
- **Power and OSNR:** The link is planned around a receiver sensitivity of −20 dBm and an operating margin greater than 6 dB. The post-amplifier OSNR design target is at least 21 dB, compared with the 17 dB threshold adopted in this report for a BER target of 10^{-12}.

2. Stage 1: Interpretation of Communication Requirements & Governing Physics

Converting the stated distance, data rate, and upgrade objective into a practical optical link requires consideration of the main propagation effects and the characteristics of the selected components:

- **1550 nm transmission band:** Silica fiber has a low-attenuation region near 1550 nm, with a typical loss of approximately 0.18–0.20 dB/km. This wavelength range also overlaps the gain band of Erbium-Doped Fiber Amplifiers (EDFAs), making it suitable for C-band optical amplification.
- **Waveguide behavior and coherence:** The fiber core is only several micrometres across, so its dimensions are comparable with the 1.55 µm optical wavelength. A ray-only model is therefore insufficient for describing diffraction, evanescent fields, and guided modes. Electromagnetic wave analysis using Maxwell's equations and the weak-guidance approximation is appropriate. A stable, narrow-linewidth DFB laser source is considered for the high-speed link.
- **Numerical aperture and normalized frequency:** The numerical aperture (NA) describes the light-acceptance capability, while the normalized frequency or V-number determines the supported modes. Keeping the V-number below the single-mode cutoff prevents higher-order modes and avoids intermodal dispersion.
- **Relevant fiber categories:** The selected G.652.D fiber is a low-water-peak single-mode type. G.655 fiber is included as a comparison because it provides non-zero, reduced dispersion. G.657.A fiber is suitable for building-entry sections because its bend-insensitive design reduces macrobending loss.
- **Attenuation sources:** Signal power is reduced by intrinsic glass effects such as Rayleigh scattering and infrared absorption, as well as by cabling-related microbending. Additional localized losses arise at connectors, fusion splices, and bends along the premises entry route.

- **Dispersion and polarization effects:** Material and waveguide dispersion cause optical pulses to broaden in time. When neighbouring pulses overlap, ISI can occur. Small core asymmetries and mechanical stresses can also create birefringence, causing the two polarization components to experience different propagation delays, known as PMD.
- **OSNR and detection quality:** Optical amplifiers add ASE noise as they increase signal power. The receiver's BER is therefore affected by the ratio of signal power to noise power. The link must retain sufficient OSNR at the receiver to meet the selected detection-performance target.

3. Stage 2: Fiber Investigation & Propagation Configuration Selection

The optical guidance properties of two step-index fiber candidates are compared at the operating wavelength of 1550 nm (1.55 µm). Their core dimensions and refractive indices are used to determine the numerical aperture and V-number. The two quantities used to assess light acceptance and modal behavior are calculated from the following expressions:

$$NA = \sqrt{(n_1^2 - n_2^2)}$$
$$V = (2\pi\, a\, /\, \lambda) \times NA$$

In these expressions, a denotes the core radius, λ is the free-space wavelength, and n_1 and n_2 are the refractive indices of the core and cladding.

Candidate Fiber	Core Radius (a)	Refractive Indices	Numerical Aperture (NA)	V-Number (1550 nm)	Guided Modes (M)
Fiber A (Step-Index)	4.0 µm	n_1 = 1.4500 n_2 = 1.4450	0.1203	1.9508	1 (Single-Mode, LP_{01} / HE_{11})
Fiber B (Step-Index)	25.0 µm	n_1 = 1.4800 n_2 = 1.4650	0.2102	21.30	≈ 227 (Multimode)

Calculated Results and Mode Assessment

1. Fiber A — Analytical Derivation

Numerical Aperture: NA = $\sqrt{(1.4500^2 - 1.4450^2)}$ = $\sqrt{(2.102500 - 2.088025)}$ = $\sqrt{0.014475}$ = 0.120312 ≈ **0.1203**

Normalized Frequency: V = (2π × 4.0 µm / 1.55 µm) × 0.120312 = (25.1327 / 1.55) × 0.120312 = 16.2146 × 0.120312 = **1.9508**

Modal Cutoff Verification: For a cylindrical step-index waveguide, the cutoff of the second-order LP_{11} mode occurs at the first zero of the J_0 Bessel function, V_c = 2.4048. Because V_A = 1.9508 < 2.405, higher-order transverse modes are cut off. Fiber A operates strictly in the fundamental single-mode regime (HE_{11} / LP_{01}), so modal dispersion is non-existent.

2. Fiber B — Analytical Derivation

Numerical Aperture: NA = $\sqrt{(1.4800^2 - 1.4650^2)}$ = $\sqrt{(2.190400 - 2.146225)}$ = $\sqrt{0.044175}$ = 0.210178 ≈ **0.2102**

Normalized Frequency: V = (2π × 25.0 µm / 1.55 µm) × 0.210178 = (157.0796 / 1.55) × 0.210178 = 101.3417 × 0.210178 = 21.2998 ≈ **21.30**

Guided Mode Count: For large-V multimode step-index fibers, M ≈ V^2 / 2 = $(21.2998)^2$ / 2 = 453.68 / 2 ≈ **227 guided modes**

Modal Dispersion Impact: Supporting ~227 propagating modes leads to differential modal group delay on the order of several nanoseconds over a 60 km span. This completely destroys a 10 Gb/s signal (bit period T_b = 100 ps), making Fiber B unviable for high-speed transmission.

Comparison of Wave-Optics and Ray-Optics Models

The suitable propagation model depends on the relationship between the core size and the optical wavelength. Candidate Fiber B has a much larger core radius (25.0 µm) than the wavelength, so geometrical ray optics provides a useful description of its propagation paths and acceptance angles. Fiber A has a smaller core radius of 4.0 µm, making wave effects important. Diffraction, field penetration into the cladding, boundary phase behavior, and polarization must therefore be considered using electromagnetic waveguide analysis.

> **Design Decision 1 – Fiber Selection —** The recommended arrangement uses ITU-T G.652.D single-mode fiber for the 60 km backbone and G.657.A fiber at the customer-premises entrance. Candidate Fiber A meets the single-mode condition because its calculated V-number (1.9508) is below 2.405. This avoids intermodal dispersion and provides the low-loss transmission path required for the 10 Gb/s service and the planned capacity upgrade.

4. Stage 3: Development & Evaluation of the Optical Power Budget

The optical power budget is evaluated using the transmitter output and the receiver sensitivity as the reference limits. The assumed launch power is +3.0 dBm, while the receiver sensitivity is −20.0 dBm. These values provide a total available loss allowance of 23.0 dB. A minimum operating reserve of 6.0 dB is retained as the design objective.

Link Parameter / Loss Mechanism	Standard G.652.D Link	Optimized G.657.A Entry + DCF Link
Distributed fiber attenuation (0.18 dB/km × 60 km)	10.80 dB	10.80 dB
Optical connectors (4 mated pairs × 0.40 dB)	1.60 dB	1.60 dB
Fusion splices (10 splices × 0.08 dB per 6 km drum)	0.80 dB	0.80 dB
Premises / building-entry macrobend loss	0.80 dB (Standard G.652.D)	0.10 dB (Bend-Insensitive G.657.A)
Dispersion-Compensating Fiber (DCF) module loss	0.00 dB (Uncompensated)	5.10 dB (10.2 km @ 0.50 dB/km)

Optical Power Budget Calculation

> **1. Baseline Uncompensated Link (with G.657.A Building Entry)**
>
> **Total Channel Attenuation:** A_{total} = 10.80 + 1.60 + 0.80 + 0.10 = **13.30 dB**
>
> **Power at Receiver:** $P_{rx} = P_{tx} - A_{total}$ = +3.0 dBm − 13.30 dB = **−10.30 dBm**
>
> **Effective System Margin:** Margin = $P_{rx} - P_{rx,sens}$ = −10.30 dBm − (−20.00 dBm) = **9.70 dB**
>
> **Compliance:** The 9.70 dB margin comfortably exceeds the 6.0 dB design requirement by 3.70 dB, providing high protection against component aging and thermal drift.
>
> **2. Fully Dispersion-Compensated Link (incorporating 10.2 km DCF Module)**
>
> **Total Channel Attenuation:** $A_{total,DCF}$ = 13.30 dB + 5.10 dB = **18.40 dB**
>
> **Power at Receiver:** P_{rx} = +3.0 dBm − 18.40 dB = **−15.40 dBm**
>
> **System Margin without Inline Gain:** Margin = −15.40 dBm − (−20.00 dBm) = **4.60 dB**
>
> **Engineering Assessment:** Under passive conditions, the 4.60 dB margin satisfies functional reception (P_{rx} > −20 dBm) but falls 1.40 dB short of the 6.0 dB design target. This is resolved by incorporating an inline optical amplifier (+14.0 dB signal gain), which elevates the power level to −2.0 dBm and increases system headroom to about 18 dB.

Main Contributors to Fiber Attenuation

For the 60 km G.652.D span, the assumed fiber attenuation is 10.80 dB in total, based on 0.18 dB/km. This distributed loss is attributed to three principal effects:

- **Rayleigh scattering (0.11 dB/km):** contributes approximately 6.60 dB, or 61.1% of the stated glass loss. It results from small-scale refractive-index variations in the silica and follows an approximate inverse-fourth-power wavelength dependence.
- **Infrared absorption (0.04 dB/km):** contributes about 2.40 dB, representing approximately 22.2% of the total fiber loss.
- **Cabling and microbending effects (0.03 dB/km):** account for about 1.80 dB, or the remaining 16.7%.

Connector, splice, and entry-bend losses are additional localized contributions. Using G.657.A at the building entrance reduces the assumed macrobending loss from 0.80 dB to 0.10 dB, a reduction of 0.70 dB.

> **Design Decision 2 – Power Budget —** A hybrid cable layout is selected, with G.652.D used along the main route and G.657.A used at the premises entry. The bend-loss allowance at the entry is thereby reduced to 0.10 dB. The calculated uncompensated received power is −10.30 dBm, giving a 9.70 dB margin. Adding the DCF section increases the total loss to 18.40 dB, so optical amplification is included to restore the required operating headroom.

5. Stage 4: Investigation & Selection of Dispersion / Polarization Management

Operation at 10 Gb/s and the proposed 100 Gb/s upgrade requires control of chromatic dispersion and polarization mode dispersion. Both effects can alter pulse timing and degrade the received waveform if they are not accounted for in the link design.

Chromatic-Dispersion Pulse-Spreading Assessment

The dispersion-induced delay spread is estimated using the following relationship:

$$\Delta\tau_{CD} = D \times \Delta\lambda \times L$$

Here, D represents the fiber dispersion coefficient, Δλ is the assumed source linewidth of 0.4 nm, and L is the 60 km fiber length. The output pulse duration is estimated by combining the input pulse width and dispersion spread using the root-sum-square (RSS) method. The assumed input pulse width is 30 ps.

$$\tau_{out} = \sqrt{(\tau_{in}^2 + \Delta\tau_{CD}^2)}$$

> **1. Standard ITU-T G.652.D Link**
>
> **Dispersion Parameter:** D = 17.0 ps/(nm·km) at 1550 nm
>
> **Delay Spread:** $\Delta\tau_{CD}$ = 17.0 × 0.4 nm × 60.0 km = **408.0 ps**
>
> **RSS Output Pulse Width:** $\tau_{out} = \sqrt{(30.0^2 + 408.0^2)} = \sqrt{(900 + 166464)} = \sqrt{167364}$ = **409.10 ps**
>
> **Impact at 10 Gb/s:** The nominal bit slot is $T_b = 1 / (10 \times 10^9)$ = 100 ps. Because the output pulse width (409.10 ps) is more than four times the bit period, the pulse spills across four adjacent bit intervals. This creates catastrophic Inter-Symbol Interference (ISI) and complete eye closure, making uncompensated transmission impossible.
>
> **2. Non-Zero Dispersion-Shifted Fiber (ITU-T G.655) — Comparative Link**
>
> **Dispersion Parameter:** D = 4.5 ps/(nm·km) at 1550 nm
>
> **Delay Spread:** $\Delta\tau_{CD}$ = 4.5 × 0.4 nm × 60.0 km = **108.0 ps**
>
> **RSS Output Pulse Width:** $\tau_{out} = \sqrt{(30.0^2 + 108.0^2)} = \sqrt{(900 + 11664)} = \sqrt{12564}$ = **112.09 ps**
>
> **Assessment:** Although G.655 reduces pulse broadening compared to G.652.D, the resulting 112.09 ps pulse width still exceeds the 100 ps bit slot, showing that dispersion management is mandatory regardless of fiber choice.

DCF Length and Compensation Design

The G.652.D span accumulates positive chromatic dispersion equal to the product of its dispersion coefficient and length. A DCF section with negative dispersion is therefore added. The required DCF length is obtained by dividing the accumulated span dispersion by the magnitude of the DCF dispersion coefficient:

$$L_{DCF} = (D_{SMF} \times L_{SMF}) \,/\, |D_{DCF}| = (17.0 \times 60.0) \,/\, 100.0 = 1020 \,/\, 100.0 = 10.20 \text{ km}$$

At the assumed attenuation of 0.50 dB/km, the 10.20 km DCF section contributes an insertion loss of 5.10 dB.

Polarization-Mode Dispersion and Birefringence

Small departures from a perfectly circular core and non-uniform mechanical stress can make the fiber birefringent. The two orthogonal polarization components of the fundamental mode may then travel at slightly different group velocities. Their relative delay is called differential group delay (DGD), and the statistical PMD estimate increases with the square root of the fiber length.

$$\Delta\tau_{PMD} = D_{PMD} \times \sqrt{L}$$

For the 60 km single-mode span, the calculation uses a PMD coefficient of 0.10 ps/√km:

$$\Delta\tau_{PMD} = 0.10 \text{ ps}/\sqrt{\text{km}} \times \sqrt{60.0 \text{ km}} = 0.10 \times 7.74597 = 0.7746 \text{ ps} \approx 0.775 \text{ ps}$$

Verification Against the PMD Design Limit

- **At 10 Gb/s:** the bit interval is 100 ps and the 10% criterion permits up to 10 ps of PMD. The calculated value of 0.775 ps is comfortably below this limit, so the expected PMD contribution is small under the assumptions used.
- **At 100 Gb/s:** the bit interval is 10 ps and the corresponding 10% limit is 1.0 ps. The estimated 0.775 ps remains below this value. Coherent systems can also use adaptive digital signal processing to compensate for polarization-related delay, providing additional tolerance to PMD.

Optical Beat Length

Birefringence is expressed as $B = |n_x - n_y|$. For the assumed standard-fiber value of 1.0×10^{-6}, the beat length at 1550 nm is calculated using the wavelength divided by birefringence:

$$L_B = \lambda \,/\, B = 1.55 \times 10^{-6} \text{ m} \,/\, 1.0 \times 10^{-6} = 1.55 \text{ m}$$

Polarization-maintaining fibers use much larger birefringence (approximately 10^{-4}), resulting in a shorter beat length of about 15 mm.

> **Design Decision 3 – Dispersion and PMD —** The proposed configuration combines G.652.D fiber with a 10.2 km DCF section having a dispersion coefficient of −100 ps/(nm·km). This is intended to offset the span's accumulated dispersion of 1020 ps/nm. The calculated PMD of 0.775 ps is within the stated limits for both the 10 Gb/s baseline and the 100 Gb/s upgrade assessment.

6. Stage 5: Evaluation of OSNR & Optical Amplification Strategy

For the amplifier assessment, OSNR is referenced to a 0.10 nm optical bandwidth, corresponding to approximately 12.5 GHz at 1550 nm. The report adopts 17 dB as the minimum OSNR criterion for the 10 Gb/s direct-detection receiver and the stated BER objective of 10^{-12}.

> **Optical Conditions Before Amplification**
>
> **Signal power:** $P_{sig,in}$ = −16.0 dBm
>
> **ASE noise power:** $P_{ase,in}$ = −39.0 dBm
>
> **Input OSNR:** $OSNR_{in}$ = −16.0 − (−39.0) = **23.0 dB**, which is 6 dB above the 17 dB reference threshold.

EDFA Operating Parameters

An inline Erbium-Doped Fiber Amplifier (EDFA) is included to compensate for the channel loss introduced by the transmission span and DCF. The assumed amplifier characteristics are:

- **Signal gain:** +14.0 dB
- **ASE noise-power increase:** +16.0 dB, including the effect of the amplifier noise figure (approximately 5.0–5.5 dB).

> **Calculated Signal and Noise Levels After Amplification**
>
> **Output signal power:** $P_{sig,out}$ = −16.0 dBm + 14.0 dB = **−2.0 dBm**
>
> **Output ASE noise power:** $P_{ase,out}$ = −39.0 dBm + 16.0 dB = **−23.0 dBm**
>
> **Output OSNR:** $OSNR_{out}$ = −2.0 − (−23.0) = **21.0 dB**
>
> **OSNR margin:** 21.0 − 17.0 = **4.0 dB**

The amplifier raises the signal level but also increases ASE noise. Under the assumed values, OSNR changes from 23.0 dB at the input to 21.0 dB at the output. This remains 4.0 dB above the adopted 17 dB requirement. The resulting signal level is −2.0 dBm; receiver compatibility and overload limits should be confirmed against the selected receiver datasheet.

> **Design Decision 4 – Amplification and OSNR —** The proposed inline EDFA is assigned a signal gain of 14 dB and an ASE noise increase of 16 dB. The calculated output OSNR is 21 dB, leaving a 4 dB margin over the report's 17 dB criterion. This provides the required noise allowance for the proposed transmission link, subject to equipment-level verification.

7. Stage 6: Verification Matrix & Final Engineering Deployment Recommendation

The following compliance table compares the principal design targets with the values obtained from the calculations in the preceding sections.

Engineering Requirement	Target Specification	Achieved Design Parameter	Compliance Status
Modal Condition & Fiber Classification	Single-mode operation (V < 2.405); ITU-T G.652.D / G.657.A	V = 1.9508 (< 2.405); G.652.D trunk + G.657.A drop	**COMPLIANT (Pass)**
Fiber & Link Attenuation Minimization	Minimize Rayleigh scattering, absorption, and macrobending	Cable: 10.80 dB (0.18 dB/km); Total link with DCF: 18.40 dB	**COMPLIANT (Pass)**
Received Optical Power & System Operating Margin	Sensitivity: −20.0 dBm; Operational margin ≥ 6.0 dB	P_{rx} = −15.4 dBm (passive) / −2.0 dBm (amp); Margin: 4.6 dB (passive) / ≈ 18 dB (boosted)	**COMPLIANT (Pass)**
Pulse Broadening & Dispersion Compensation	Maintain pulse integrity within bit slot (τ < 100 ps)	10.2 km DCF neutralizes +1020 ps/nm; Pulse broadening eliminated	**COMPLIANT (Pass)**
PMD Performance & Upgrade Compatibility	PMD ≤ 0.10 T_b (≤ 10 ps at 10G; ≤ 1.0 ps at 100G)	Calculated PMD = 0.775 ps (< 1.0 ps single-carrier threshold)	**COMPLIANT (Pass)**
Optical Signal-to-Noise Ratio (OSNR)	Standard required OSNR ≥ 17.0 dB	Achieved OSNR = 21.0 dB (4.0 dB safety margin)	**COMPLIANT (Pass)**

Final Deployment Assessment

OVERALL DESIGN STATUS: PROPOSED FOR FIELD IMPLEMENTATION, SUBJECT TO DETAILED EQUIPMENT VALIDATION

The calculations indicate that the proposed 60 km architecture can support the specified 10 Gb/s service using a single-mode G.652.D backbone and G.657.A building-entry fiber. The design analysis gives a V-number of 1.9508, a reduced entry-bend loss of 0.10 dB, a 10.2 km DCF section to address the accumulated chromatic dispersion, and an estimated PMD of 0.775 ps. With the assumed EDFA parameters, the output OSNR is 21 dB, which is 4 dB above the adopted 17 dB criterion. These results provide a basis for the proposed link design and its potential future migration to coherent 100 Gb/s equipment.